

```
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-100-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Thu Mar  5 07:31:54 AM UTC 2026

System load:  0.39          Processes:            113
Usage of /:   25.8% of 24.44GB Users logged in:      0
Memory usage: 5%           IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,504 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 c-n-f Metadata [30.5 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/restricted amd64 c-n-f Metadata [416 B]
Get:8 http://archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8,328 B]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,805 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [332 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.7 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,747 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [632 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:18 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,564 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [318 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [32.9 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [7,044 B]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [496 B]
Get:25 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,300 B]
Get:26 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:27 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:28 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:29 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:30 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1,444 B]
Get:31 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:32 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:33 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [241 kB]
Get:34 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:35 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.1 kB]
Get:36 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,599 kB]
Get:37 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [600 kB]
Get:38 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:39 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [536 B]
Get:40 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [975 kB]
Get:41 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [218 kB]
Get:42 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [20.6 kB]
Get:44 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6,732 B]
Get:45 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:46 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Fetched 15.0 MB in 12s (1,221 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@pooja:~#
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```
root@pooja:~#
root@pooja:~# apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```

root@pooja:~# apt install -y apt-transport-https ca-certificates curl gpg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20240203).
ca-certificates set to manually installed.
gpg is already the newest version (2.4.4-2ubuntu17.4).
gpg set to manually installed.
The following NEW packages will be installed:
  apt-transport-https
The following packages will be upgraded:
  curl libcurl13t64-gnutls libcurl14t64
3 upgraded, 1 newly installed, 0 to remove and 21 not upgraded.
Need to get 906 kB of archives.
After this operation, 36.9 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3,970 B]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.7 [227 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl14t64 amd64 8.5.0-2ubuntu10.7 [342 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl13t64-gnutls amd64 8.5.0-2ubuntu10.7 [334 kB]
Fetched 906 kB in 5s (178 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 87509 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Preparing to unpack .../curl_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl14t64_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking libcurl14t64:amd64 (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Preparing to unpack .../libcurl13t64-gnutls_8.5.0-2ubuntu10.7_amd64.deb ...
Unpacking libcurl13t64-gnutls:amd64 (8.5.0-2ubuntu10.7) over (8.5.0-2ubuntu10.6) ...
Setting up apt-transport-https (2.8.3) ...
Setting up libcurl14t64:amd64 (8.5.0-2ubuntu10.7) ...
Setting up libcurl13t64-gnutls:amd64 (8.5.0-2ubuntu10.7) ...
Setting up curl (8.5.0-2ubuntu10.7) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@pooja:~#

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```

root@pooja:~# mkdir -p /etc/apt/keyrings
root@pooja:~# curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.29/deb/Release.key | gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
curl: (22) The requested URL returned error: 403
gpg: no valid OpenPGP data found.
root@pooja:~# sudo apt install -y containerd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  runc
The following NEW packages will be installed:
  containerd runc
0 upgraded, 2 newly installed, 0 to remove and 21 not upgraded.
Need to get 47.2 MB of archives.
After this operation, 178 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 containerd amd64 1.3.3-0ubuntu1~24.04.3 [8,815 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.7.28-0ubuntu1~24.04.2 [38.4 MB]
Fetched 47.2 MB in 22s (2,106 kB/s)
Selecting previously unselected package runc.
(Reading database ... 87513 files and directories currently installed.)
Preparing to unpack .../runc_1.3.3-0ubuntu1~24.04.3_amd64.deb ...
Unpacking runc (1.3.3-0ubuntu1~24.04.3) ...
Selecting previously unselected package containerd.
Preparing to unpack .../containerd_1.7.28-0ubuntu1~24.04.2_amd64.deb ...
Unpacking containerd (1.7.28-0ubuntu1~24.04.2) ...
Setting up runc (1.3.3-0ubuntu1~24.04.3) ...
Setting up containerd (1.7.28-0ubuntu1~24.04.2) ...
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

```

```

    root_path = ""
    sync_remove = false
    upperdir_label = false

[plugins."io.containerd.snapshotter.v1.zfs"]
    root_path = ""

[plugins."io.containerd.tracing.processor.v1.otlp"]

[plugins."io.containerd.transfer.v1.local"]
    config_path = ""
    max_concurrent_downloads = 3
    max_concurrent_uploaded_layers = 3

[[plugins."io.containerd.transfer.v1.local".unpack_config]]
    differ = "walking"
    platform = "linux/amd64"
    snapshotter = "overlayfs"

[proxy_plugins]

[stream_processors]

[stream_processors."io.containerd.ocicrypt.decoder.v1.tar"]
    accepts = ["application/vnd.oci.image.layer.v1.tar+encrypted"]
    args = ["--decryption-keys-path", "/etc/containerd/ocicrypt/keys"]
    env = ["OCICRYPT_KEYPROVIDER_CONFIG=/etc/containerd/ocicrypt/ocicrypt_keyprovider.conf"]
    path = "ctd-decoder"
    returns = "application/vnd.oci.image.layer.v1.tar"

[stream_processors."io.containerd.ocicrypt.decoder.v1.tar.gzip"]
    accepts = ["application/vnd.oci.image.layer.v1.tar+gzip+encrypted"]
    args = ["--decryption-keys-path", "/etc/containerd/ocicrypt/keys"]
    env = ["OCICRYPT_KEYPROVIDER_CONFIG=/etc/containerd/ocicrypt/ocicrypt_keyprovider.conf"]
    path = "ctd-decoder"
    returns = "application/vnd.oci.image.layer.v1.tar+gzip"

[timeouts]
    "io.containerd.timeout.bolt.open" = "0s"
    "io.containerd.timeout.metrics.shimstats" = "2s"
    "io.containerd.timeout.shim.cleanup" = "5s"
    "io.containerd.timeout.shim.load" = "5s"
    "io.containerd.timeout.shim.shutdown" = "3s"
    "io.containerd.timeout.task.state" = "2s"

[tttrpc]
    address = ""
    gid = 0
    uid = 0
root@pooja:~#

```

```

root@pooja:~# sudo systemctl restart containerd
root@pooja:~# sudo systemctl enable containerd
root@pooja:~# apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
21 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

```

pooja@pooja:~$ sudo -i
[sudo] password for pooja:
root@pooja:~# apt install -y apt-transport-https ca-certificates curl gpg
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
ca-certificates is already the newest version (20240203).
curl is already the newest version (8.5.0-2ubuntu10.7).
curl set to manually installed.
gpg is already the newest version (2.4.4-2ubuntu17.4).
0 upgraded, 0 newly installed, 0 to remove and 21 not upgraded.
root@pooja:~# mkdir -p /etc/apt/keyrings
root@pooja:~# curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.29/deb/Release.key | gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
File '/etc/apt/keyrings/kubernetes-apt-keyring.gpg' exists. Overwrite? (y/N) y
root@pooja:~# echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /" | tee /etc/apt/sources.list.d/kubernetes.list
deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v1.29/deb/ /
root@pooja:~# sudo apt install -y containerd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
containerd is already the newest version (1.7.28-0ubuntu1~24.04.2).
0 upgraded, 0 newly installed, 0 to remove and 21 not upgraded.
root@pooja:~#

```

no VM guests are running outdated hypervisor (qemu) binaries on this host.

```

root@pooja:~# apt-mark hold kubelet kubeadm kubectl
kubelet set on hold.
kubeadm set on hold.
kubectl set on hold.
root@pooja:~# sudo swapoff -a
root@pooja:~# free -m

```

	total	used	free	shared	buff/cache	available
Mem:	3915	407	2720	1	1013	3508
Swap:	0	0	0			

```

root@pooja:~#

```